

KKK OIL FACTORY

Enterprise ERP & GST Billing Suite

Standard GST Operations, RBAC & Path-by-Path Manual

OPERATIONAL DOCUMENT SPECIFICATION

System Version: 2.5.0 Production Release
Enterprise: KKK Oil Factory, Tamil Nadu, India
Target Roles: Super Admin, Admin, Manager, Cashier, Supervisor, Driver
Supported Environments: Web Browser (Clean URL), Android Tablet/Phone, iOS, Desktop
Classification: Industrial ERP, Expeller Manufacturing, Statutory GST Billing

Manual Scope & Architecture

This manual provides complete operational and technical documentation for the KKK Oil Factory Enterprise ERP. It details every system path (/login, /dashboard, /billing, /counter, /products, /inventory, /production, /purchase, /logistics, /reports, /employees, /settings), every form field, every button action, and step-by-step operating workflows for factory floor and billing staff.

Role-Based Security Model

- Super Admin: Full system executive oversight, pricing approval, and user administration.
- ERP Administrator: 15 core modules, master catalog, customer accounts, and company settings.
- Plant / Shift Manager: Cashier monitoring, expeller yield approval, customer credit, and purchase orders.
- Billing Cashier: High-speed POS counter billing, B2B tax invoices, and thermal receipt printing.
- Production Supervisor: Machine run logging, seed input, oil recovery, and byproduct cake bagging.
- Logistics Driver: Dispatch gate passes, delivery routes, and digital Proof of Delivery (POD).
- Warehouse Incharge: Silo dip stick measurements, packaged stock, and transfer notes.

1. Complete Path-by-Path System Manual

Authentication Gate: /login

Authenticates staff members, decrypts authorization tokens, and initializes reactive state. Features Username and Password fields with an eye visibility toggle, 'Sign In to ERP System' button, and quick demo switchers for Super Admin, Admin, Manager, and Cashier.

Executive Dashboard: /dashboard

Real-time operational cockpit providing Today's Gross Sales, Cash Balance, Silo Dip Levels, and Low Stock alerts. Action buttons include '+ New B2B Invoice', '+ Quick POS Counter', '+ Start Milling Batch', '+ Record Seed Inward', and '+ Delivery Gate Pass'.

Standard B2B GST Invoicing: /billing (erp_billing)

Generates formal tax invoices compliant with Rule 46 of CGST Rules 2017. Header controls include Customer Selector (shows credit balance), Route dropdown, Godown selector, Pricing tier radio (Retail, Semi-Wholesale, Bulk Wholesale), Payment Mode, and Payment Status.

- Item Search: Real-time product search by code or name; auto-fills HSN and rate.
- Tax Calculation: Compares customer GSTIN with state 33. Intra-state: CGST 2.5% + SGST 2.5% = 5%. Inter-state: IGST 5%.
- Bottom Tabs: Tab 0 (Dispatch details, PO, Vehicle No), Tab 1 (E-Way Bill No), Tab 2 (E-Invoice IRN), Tab 3 (Tax ledgers).
- Actions: 'Save Invoice' (deducts stock, updates ledger), 'Print A4 Tax Invoice', 'Print Thermal Receipt', 'Clear Form'.

Billing History & Cancellation: /billing-history

Audit repository of all issued GST tax invoices. Features search by invoice number or customer, payment filter chips (All, Paid, Credit, Partial), 'Reprint A4' button, 'Reprint Thermal' button, and 'Cancel Bill' button (prompts for reason, restores stock to godown, and reverses customer debit).

2. Manufacturing, Procurement & Stock

High-Velocity POS Counter Billing: /counter

Designed for walk-in retail factory outlets. Features one-tap product catalog grid, oil category filter chips, pricing tier selector, active cart steppers (+/-), cash tendered calculator with change return, and 'Complete & Print Thermal' button.

Master Product Catalog: /products

Master registry for all oils and byproducts. Header buttons: '+ Add New Product Master' and 'Export Master CSV'. Add/Edit modal configures basic specs, tier rates (Agency, Wholesale, Retail, MRP), initial stock, reorder thresholds, and HSN codes (1508/1515/2306).

Warehouse Inventory & Silo Tanks: /inventory

Monitors bulk crude and filtered oil tanks using calibrated dip stick measurements (cm to Liters). Packaged stock grid shows finished tins, bottles, and pouches with color-coded stock alerts. Actions include 'Adjust Physical Stock' reconciliation and 'Stock Transfer Note'.

Seed Procurement & Weighbridge: /purchase

Records raw seed intake from mandis. '+ Record Seed & Raw Material Inward' opens Goods Receipt Note (GRN) dialog capturing supplier, weighbridge net weight, intake rate (Rs./kg), moisture deduction, and QC Status ('Passed' or 'Pending Lab Test').

Expeller Production Milling: /production

Tracks expeller crushing batches. '+ Start New Production Milling Run' records planned seed input (kg), filtered oil output (Liters), oil cake byproduct (kg), and wastage. Automatically computes extraction yield percentage.

3. Accounts, Logistics & Daily SOPs

Customer & Supplier Masters: /customers & /suppliers

Customer screen manages Regular/Retail vs Dealer categories, GSTIN validation, credit limits, and ledger statements. Supplier screen tracks mandi vendors, packaging suppliers, and accounts payable.

Logistics Fleet & Gate Passes: /logistics

'+ Generate Dispatch Gate Pass' links vehicle registration (e.g. TN 28 BK 5521), driver, route, and invoice. Provides trip status stepper ('Dispatched' -> 'In Transit' -> 'Delivered') and mobile Proof of Delivery (POD) confirmation.

Tax Analytics & Compliance: /reports

Displays total gross turnover, taxable base, and output GST. Breaks down B2B supplies (Table 4) vs B2C supplies (Table 7), CGST/SGST/IGST splits, and HSN Table 12 summary. Actions: 'Export GSTR-1 Excel/CSV' and 'Download Daily Daybook PDF'.

Employee Administration & Settings: /employees & /settings

Employee screen manages staff accounts, role assignment, active/inactive switches, and the 15-module RBAC matrix. Settings screen configures company trade name, registered address, GSTIN, FSSAI number, bank account/IFSC, and default printer formats.

CORE CASHIER BILLING SOP

1. Morning Sign-In: Sign in at /login with cashier credentials.
2. Access /counter: Verify opening cash drawer balance.
3. Ring Up Items: Tap product cards; adjust quantities using steppers.
4. Collect Payment: Select Cash or UPI; enter tendered cash to calculate change.
5. Print & Handover: Click 'Complete & Print Thermal'; hand receipt and pack to buyer.
6. Shift Closing: Check /billing-history and submit physical cash to Manager.